COMPUTER NETWORKS AND PROTOCOLS (EC363IA) – PBL SEE

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Problem Statement : Traditional Wi-Fi bandwidth allocation uses fixed rules and cannot adapt to changing network conditions. This results in increased delay, packet loss, and reduced throughput during network congestion. Therefore, an intelligent system is needed to dynamically predict network priority and allocate bandwidth efficiently using Machine Learning.

Introduction

Wireless Local Area Networks (Wi-Fi) have become an essential part of modern communication, connecting a large number of devices such as smartphones, laptops, and IoT devices. As the number of connected users increases, the network often experiences congestion, resulting in higher delay, increased packet loss, and reduced throughput, which affect the Quality of Service (QoS). Traditional bandwidth allocation methods rely on fixed rules and are unable to adapt to changing network conditions. To address this problem, this project proposes a Machine Learning-based adaptive Wi-Fi bandwidth allocation system that predicts network priority using Delay, Packet Loss, and Throughput. The system compares the performance of Decision Tree and Random Forest algorithms, and based on the predicted priority, recommends appropriate bandwidth allocation to improve overall network performance and user experience.

Objectives

- Develop an adaptive Wi-Fi bandwidth allocation system using Machine Learning.
- Predict network priority (High, Medium, and Low) based on Delay, Packet Loss, and Throughput.
- Compare the performance of Decision Tree and Random Forest algorithms for QoS prediction.
- Design an interactive Streamlit dashboard for real-time prediction, visualization, and performance analysis.

Methodology

1. Network Parameter Collection

Real-time Wi-Fi metrics captured:
Delay (ms), Packet Loss (%), Throughput (Mbps)

2. Synthetic Dataset Generation

2000 labelled samples generated using
normalised composite scoring formula

3. QoS Priority Labelling

Score = (Delay/200) + (PacketLoss/20) − (Throughput/100)
High · Medium · Low priority assigned per threshold

4. Feature Selection & Dataset Design

3 input features structured into training matrix;
QoS priority labels assigned as target variable

5. Model Training

Decision Tree and Random Forest classifiers
trained on 75% of dataset (stratified split)

6. Performance Evaluation

Evaluated using Accuracy, Precision, Recall,
F1-Score and 5-fold Cross-Validation

7. Best Performing Model (Random Forest)

RF achieves 94.60% accuracy vs DT's 89.40%;
selected as the adaptive QoS prediction engine

8. Adaptive QoS Priority Prediction

Trained RF model predicts priority class
for incoming real-time network conditions

9. Bandwidth Allocation Decision

Priority mapped to Wi-Fi queue policy:
High → 60–80% · Medium → 20–35% · Low → 5–15%

10. Adaptive Wi-Fi QoS Provisioning

Dynamic queue assignment enforced:
Priority Queue · Fair Queue · Best-Effort Queue

Results and Performance Analysis



Metric	Decision Tree	Random Forest	Δ (RF − DT)
	— Overall —		
Test Accuracy	0.8940	0.9460	+0.0520
CV Accuracy (mean ± std)	0.8710 ± 0.0115	0.9290 ± 0.0174	+0.0580
	— Macro Averages —		
Macro Avg Precision	0.8848	0.9399	+0.0552
Macro Avg Recall	0.8770	0.9371	+0.0601
Macro Avg F1-Score	0.8804	0.9385	+0.0580
	— Weighted Averages —		
Weighted Avg Precision	0.8945	0.9458	+0.0513
Weighted Avg Recall	0.8940	0.9460	+0.0520
Weighted Avg F1-Score	0.8938	0.9459	+0.0520
	— Per-Class F1-Score —		
High Priority F1-Score	0.9064	0.9489	+0.0425
Medium Priority F1-Score	0.7938	0.8941	+0.1003
Low Priority F1-Score	0.9412	0.9724	+0.0312
	— Per-Class Precision —		
High Priority Precision	0.9380	0.9559	+0.0179
Medium Priority Precision	0.7907	0.8976	+0.1069
Low Priority Precision	0.9256	0.9662	+0.0406

Tools and Techniques Used

Tools
Python 3.x, Visual Studio Code (VS Code), Streamlit, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Techniques
Synthetic Dataset Generation, Feature Engineering, Decision Tree Classification, Random Forest Classification, QoS Priority Prediction, Adaptive Bandwidth Allocation, Train-Test Split (80:20), Cross-Validation, Performance Evaluation (Accuracy, Precision, Recall, F1-Score), Real-Time Dashboard Visualization

References

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